

```
create database lab9;
use lab9;
```

```
-- Table: Students
```

```
CREATE TABLE Students (
  student_id INT PRIMARY KEY,
  student_name VARCHAR(50),
  age INT,
  major VARCHAR(50)
);
```

```
-- Table: Courses
```

```
CREATE TABLE Courses (
  course_id INT PRIMARY KEY,
  course_name VARCHAR(50),
  instructor VARCHAR(50),
  credits INT
);
```

```
-- Table: Enrollments
```

```
CREATE TABLE Enrollments (
  enrollment_id INT PRIMARY KEY,
  student_id INT,
  course_id INT,
  grade CHAR(1),
  FOREIGN KEY (student_id) REFERENCES Students(student_id),
  FOREIGN KEY (course_id) REFERENCES Courses(course_id)
);
```

```
-- Insert Data:
```

```
-- Insert into Students
```

```
INSERT INTO Students (student_id, student_name, age, major) VALUES
(1, 'Alice', 21, 'Computer Science'),
(2, 'Bob', 22, 'Mathematics'),
(3, 'Charlie', 20, 'Physics'),
(4, 'Diana', 23, 'Engineering'),
(5, 'Eve', 21, 'Business');
```

```
select * from Students;
```

```
-- Insert into Courses
```

```
INSERT INTO Courses (course_id, course_name, instructor, credits) VALUES
(101, 'Database Systems', 'Dr. Smith', 3),
(102, 'Calculus I', 'Dr. Johnson', 4),
(103, 'Operating Systems', 'Dr. Brown', 3),
(104, 'Microeconomics', 'Dr. Green', 3),
(105, 'Linear Algebra', 'Dr. Johnson', 4);
```

```
-- Insert into Enrollments
```

INSERT INTO Enrollments (enrollment\_id, student\_id, course\_id, grade) VALUES

(1, 1, 101, 'A'),

(2, 1, 103, 'B'),

(3, 2, 102, 'A'),

(4, 3, 101, 'B'),

(5, 3, 105, 'C'),

(6, 4, 104, 'A'),

```
(7, 5, 105, 'B');
```

Students Table:

```
26 );  
27 -- Insert Data:  
28 -- Insert into Students  
29 INSERT INTO Students (student_id, student_name, age, major) VALUES  
30 (1, 'Alice', 21, 'Computer Science'),  
31 (2, 'Bob', 22, 'Mathematics'),  
32 (3, 'Charlie', 20, 'Physics'),  
33 (4, 'Diana', 23, 'Engineering'),  
34 (5, 'Eve', 21, 'Business');  
35  
36 select * from Students;  
37
```

100% 24:36

Result Grid Filter Rows: Search Edit: Export/Import:

| student_... | student_na... | age  | major            |
|-------------|---------------|------|------------------|
| 1           | Alice         | 21   | Computer Science |
| 2           | Bob           | 22   | Mathematics      |
| 3           | Charlie       | 20   | Physics          |
| 4           | Diana         | 23   | Engineering      |
| 5           | Eve           | 21   | Business         |
| NULL        | NULL          | NULL | NULL             |

Students 1

- Courses Table:

100% 23:46



🔍 Search



|           | course_id | course_name       | instructor  | credits |  |
|-----------|-----------|-------------------|-------------|---------|--|
|           | 101       | Database Systems  | Dr. Smith   | 3       |  |
|           | 102       | Calculus I        | Dr. Johnson | 4       |  |
|           | 103       | Operating Systems | Dr. Brown   | 3       |  |
|           | 104       | Microeconomics    | Dr. Green   | 3       |  |
|           | 105       | Linear Algebra    | Dr. Johnson | 4       |  |
|           | NULl      | NULl              | NULl        | NULl    |  |
|           |           |                   |             |         |  |
|           |           |                   |             |         |  |
|           |           |                   |             |         |  |
|           |           |                   |             |         |  |
|           |           |                   |             |         |  |
|           |           |                   |             |         |  |
|           |           |                   |             |         |  |
|           |           |                   |             |         |  |
|           |           |                   |             |         |  |
|           |           |                   |             |         |  |
|           |           |                   |             |         |  |
|           |           |                   |             |         |  |
| Courses 2 |           |                   |             |         |  |

```
48 -- Insert into Enrollments
49 INSERT INTO Enrollments (enrollment_id, student_id, course_id, grade) VALUES
50 (1, 1, 101, 'A'),
51 (2, 1, 103, 'B'),
52 (3, 2, 102, 'A'),
53 (4, 3, 101, 'B'),
54 (5, 3, 105, 'C'),
55 (6, 4, 104, 'A'),
56 (7, 5, 105, 'B');
57
58 select * from Enrollments;
```

100% 27:58

Result Grid Edit: Export/Import:

| enrollment_id | student_... | course_id | grade |
|---------------|-------------|-----------|-------|
| 1             | 1           | 101       | A     |
| 2             | 1           | 103       | B     |
| 3             | 2           | 102       | A     |
| 4             | 3           | 101       | B     |
| 5             | 3           | 105       | C     |
| 6             | 4           | 104       | A     |
| 7             | 5           | 105       | B     |
| NULL          | NULL        | NULL      | NULL  |

Enrollments 3

1.

## Inner Join

: Write a query to retrieve a list of students and the courses they are enrolled in. Display the student's name, course name, and grade.

```
select S.student_name, C.course_name, E.grade from Students S
inner join Enrollments E
ON S.student_id = E.student_id
inner join Courses C
ON E.course_id = C.course_id;
```

```
61 select S.student_name, C.course_name, E.grade from Students S
62 inner join Enrollments E
63 ON S.student_id = E.student_id
64 inner join Courses C
65 ON E.course_id = C.course_id;
66
```

100%  30:65

Result Grid

 

Filter Rows:

 Search

Export: 

| student_na... | course_name       | grade |  |
|---------------|-------------------|-------|--|
| Alice         | Database Systems  | A     |  |
| Alice         | Operating Systems | B     |  |
| Bob           | Calculus I        | A     |  |
| Charlie       | Database Systems  | B     |  |
| Charlie       | Linear Algebra    | C     |  |
| Diana         | Microeconomics    | A     |  |
| Eve           | Linear Algebra    | B     |  |

Result 4

```
select S.student_name, C.course_name from Students S
left join Enrollments E
ON S.student_id = E.student_id
left join Courses C
ON E.course_id = C.course_id;
```

```

53      (4, 3, 101, 'B'),
54      (5, 3, 105, 'C'),
55      (6, 4, 104, 'A'),
56      (7, 5, 105, 'B');
57
58 •   select * from Enrollments;
59
60
61 •   select S.student_name, C.course_name, E.grade from Students S
62       inner join Enrollments E
63       ON S.student_id = E.student_id
64       inner join Courses C
65       ON E.course_id = C.course_id;
66
67 •   select S.student_name, C.course_name from Students S
68       left join Enrollments E

```

100% 30:71

Result Grid Filter Rows: Search Export:

| student_na... | course_name       |
|---------------|-------------------|
| Alice         | Database Systems  |
| Alice         | Operating Systems |
| Bob           | Calculus I        |
| Charlie       | Database Systems  |
| Charlie       | Linear Algebra    |
| Diana         | Microeconomics    |
| Eve           | Linear Algebra    |
|               |                   |
|               |                   |
|               |                   |

Result 5

4.

### Full Outer Join

: Write a query to display all students and courses, whether or not a student is enrolled in a course. Show student names and course names.

```

select s.student_name, c.course_name from students s
left join enrollments e on s.student_id = e.student_id
left join courses c on e.course_id = c.course_id
union
select s.student_name, c.course_name
from students s
right join Enrollments e on s.student_id = e.student_id
right join courses c on e.course_id = c.course_id;

```

00% 1:79 1 error found



Filter Rows:

[illegible]

Q.5

5.

## Self Join

: Write a query to find students who are enrolled in more than one course.

Display their names and the courses they are enrolled in.

```
select s.student_name, count(c.course_id) AS total_courses, group_concat(c.course_name
separator ',') as courses_enrolled
from students s
join enrollments e on s.student_id = e.student_id
join courses c on e.course_id = c.course_id
```

```
group by s.student_id
having count(c.course_id) > 1;
```

```
-- another way
98 select s.student_name, count(c.course_id) AS total_courses, group_concat(c.course_name separ
99 from students s
100 join enrollments e on s.student_id = e.student_id
101 join courses c on e.course_id = c.course_id
102 group by s.student_id
103 having count(c.course_id)> 1;
```

100% 31:103

**Result Grid**

| student_na... | total_cours... | courses_enrolled                    |
|---------------|----------------|-------------------------------------|
| Alice         | 2              | Database Systems, Operating Systems |
| Charlie       | 2              | Database Systems, Linear Algebra    |

Export:

Result 9

6.

## Cross Join

: Write a query that retrieves all possible student-course combinations.

Display the student's name and course name.

```
select s.student_name, c.course_name
from students s
cross join courses c;
```

```

98
99     join enrollments e on s.student_id = e.student_id
100     join courses c on e.course_id = c.course_id
101     group by s.student_id
102     having count(c.course_id) > 1;
103
104 • select s.student_name, c.course_name
105     from students s
106     cross join courses c;

```

100% 22:106

Result Grid



Filter Rows:



Search

Export:



| student_na... | course_name       |  |
|---------------|-------------------|--|
| Eve           | Database Systems  |  |
| Diana         | Database Systems  |  |
| Charlie       | Database Systems  |  |
| Bob           | Database Systems  |  |
| Alice         | Database Systems  |  |
| Eve           | Calculus I        |  |
| Diana         | Calculus I        |  |
| Charlie       | Calculus I        |  |
| Bob           | Calculus I        |  |
| Alice         | Calculus I        |  |
| Eve           | Operating Systems |  |
| Diana         | Operating Systems |  |
| Charlie       | Operating Systems |  |
| Bob           | Operating Systems |  |
| Alice         | Operating Systems |  |
| Eve           | Microeconomics    |  |

Result 15